RiceGuard

Healthy Rice Leaf Identification
Documentary of the interactive notebook

Source notebook: `notebooks/rice_healthy_leaf_simulation.ipynb`
Generated: August 19, 2026
Result on this sample: **HEALTHY** (97.7% confidence)
Engine used: production model

Step 1 - Original leaf photo (3048x4064 px)



Sample used in this documentary: a healthy green rice leaf.

1. How the system identifies a healthy leaf

The RiceGuard web app does not guess a disease from a red overlay. It runs a stacked pipeline. This notebook documents **Layer 1** — the health decision — on a leaf that has no disease spots.

Layer	Question	If the answer is…
0 Rice-leaf gate	Is this a rice leaf photo?	No → stop (not rice)
1 Health model	Healthy or diseased?	Healthy → stop (this notebook)
2 Disease model	Which disease?	Only runs if Layer 1 says diseased
3 Recommendation	What should the farmer do?	PhilRice-aligned guidance

The production health model (`cnn_health.pkl` / `rf_health.pkl`) is trained on the **whole leaf**. A green blade with vein texture is still healthy. A naïve HSV “lesion” mask often paints veins red and would wrongly say brown spot or leaf smut — that is a teaching overlay, not the diagnosis.

This Jupyter kernel may be older than the trained sklearn pickle. When the pickle cannot load, the notebook uses a **Layer 1 simulation** with the same idea: compare green tissue vs brown/gray damage across the entire leaf mask. On this sample both routes agree: **HEALTHY**.

Notebook walk-through

- Step 1 — Load the photo.
- Step 2 — Letterbox-resize to 128×128 (same size the app uses).
- Step 3 — Inspect RGB / HSV colour (healthy green band).
- Step 4 — HSV lesion search (may false-positive on veins).
- Step 5 — Turn the leaf into numbers (GLCM, HSV moments, LBP, shape).
- Step 6 — Layer 1: healthy vs diseased (the decision).
- Step 7–8 — Flow diagram and farmer dashboard.

2. Step 1 — Load the healthy leaf photo

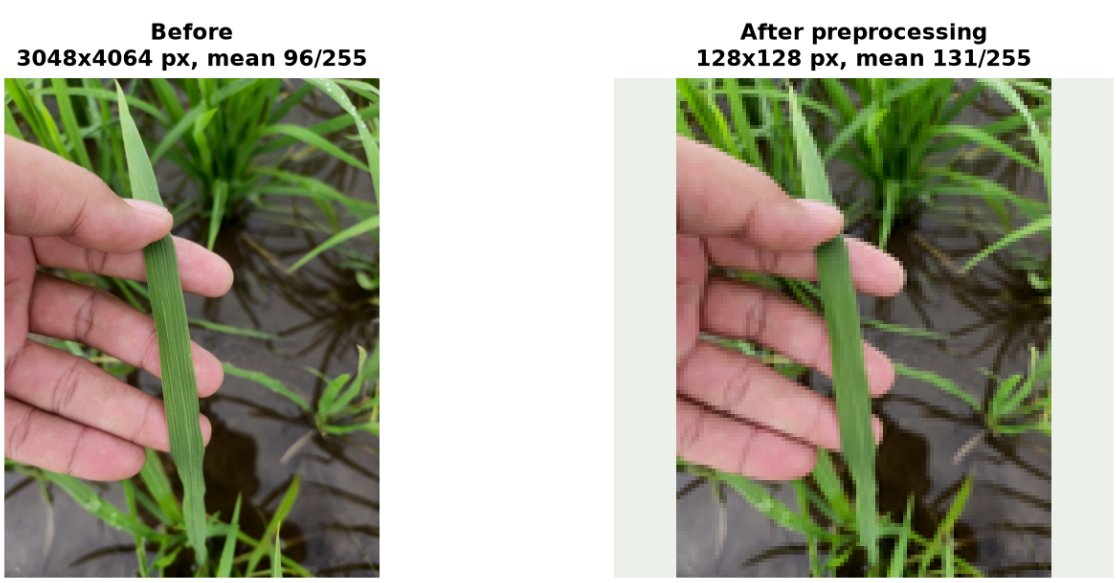
The input is a close-up of a green rice leaf with no brown or gray disease spots. Everything later is computed from this single image.



c05 01

3. Step 2 — Preprocess to 128×128

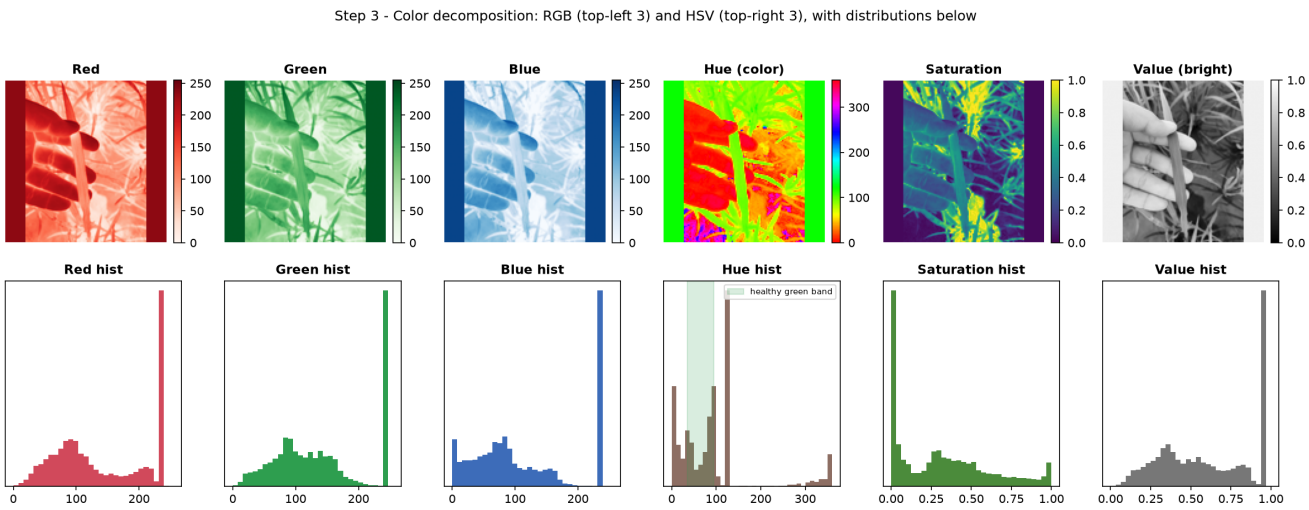
The app letterbox-resizes every upload to 128×128 with a neutral green-gray backdrop and a gentle brightness normalize. Training and inference share this step so features stay comparable.



c07 02

4. Step 3 — Colour analysis (RGB and HSV)

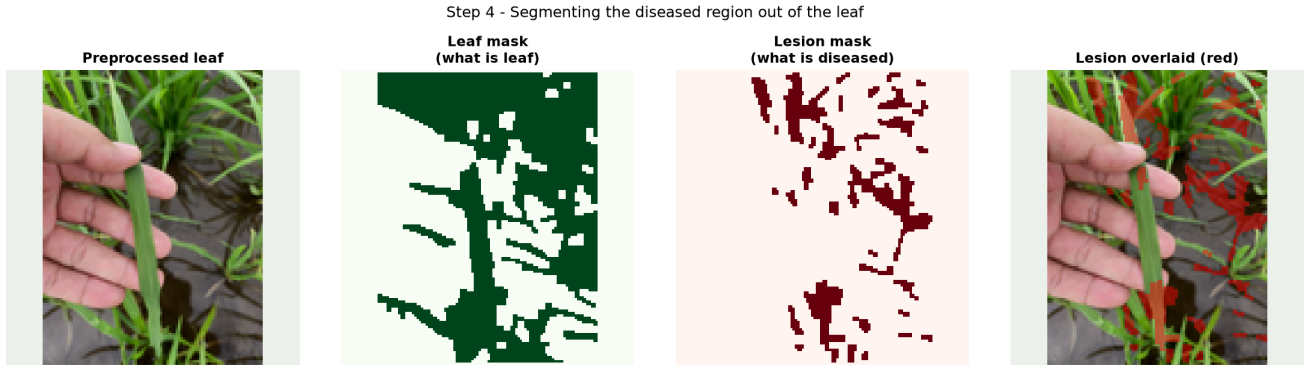
Healthy rice tissue sits in a green hue band (about 35–110°) with useful saturation. Disease pulls pixels toward brown, yellow, or gray. The histograms below are how you see that before any classifier runs.



c09 03

5. Step 4 — Search for lesions (teaching overlay)

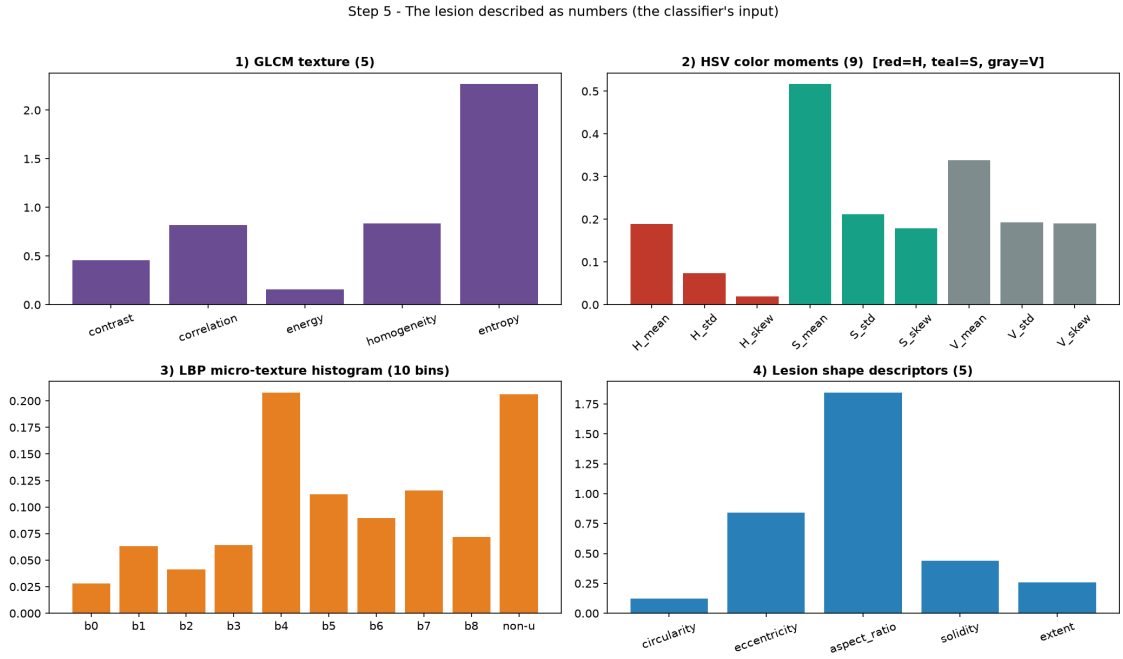
HSV rules mark off-green or dark pixels inside the leaf. On a healthy leaf this mask is often noisy: veins, shadows, and a white background can look like lesions. On this run HSV coverage was **28.5%** of the leaf — that number is **not** the diagnosis. Layer 1 looks at the whole leaf instead.



c11 04

6. Step 5 — Features (numbers the classifier sees)

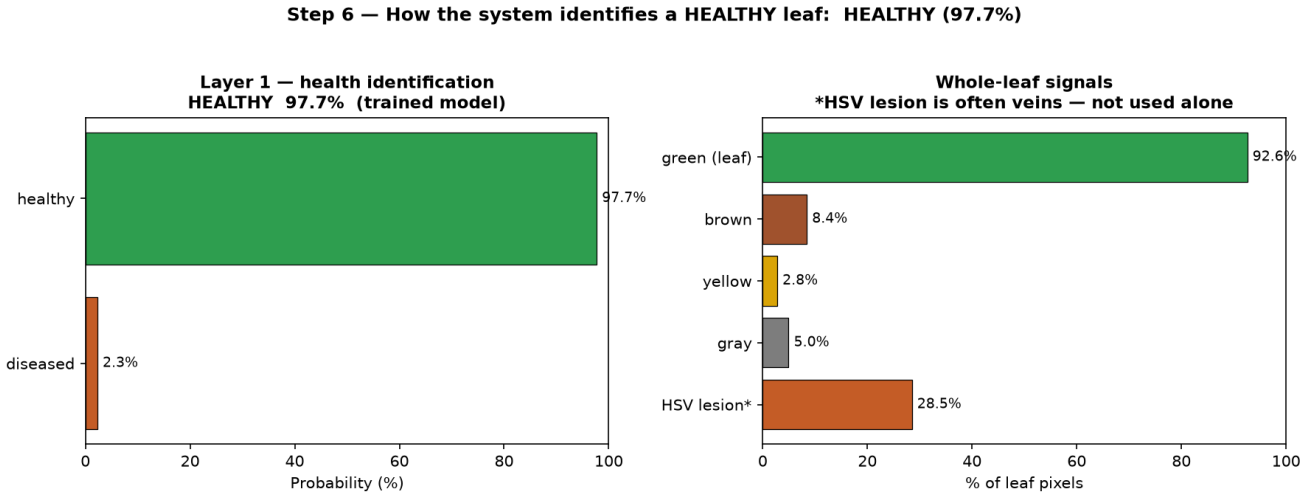
Texture (GLCM, LBP), colour moments, and shape descriptors summarise the region. The production model uses a long feature vector (or a CNN embedding). The bar charts show the teaching version of those families.



c13 05

7. Step 6 — Layer 1 health identification

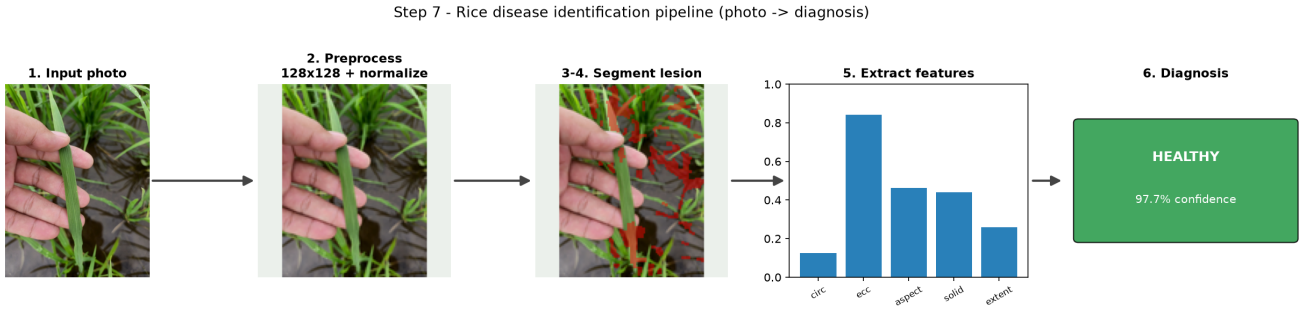
This is the decision the app makes first: **healthy** or **diseased**. Green fraction of the leaf is high; brown/yellow/gray are low. HSV lesion coverage is only a weak extra signal because it false-positives on veins. Result: **HEALTHY** at **97.7%** confidence (production model). Layer 2 (disease name) is skipped.



c15 06

8. Step 7 — End-to-end flow

Photo → preprocess → colour/lesion view → features → Layer 1 card. For a healthy leaf the path stops at HEALTHY.



c17 07

9. Step 8 — Diagnosis dashboard

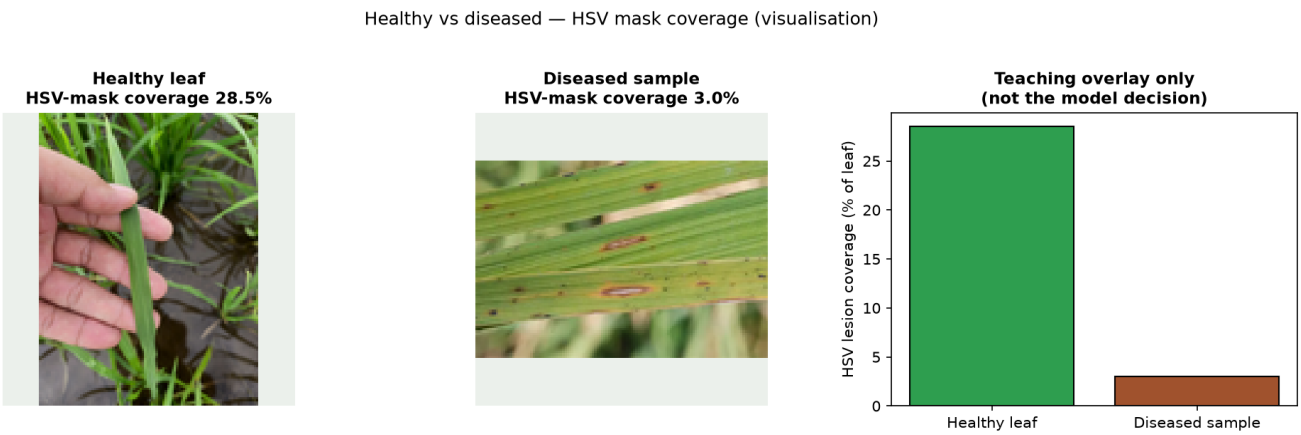
The summary a farmer would see: input photo, teaching overlay, Layer 1 bars, and PhilRice-aligned prevention tips for a healthy crop.



c19 08

10. Comparison (healthy vs a diseased sample)

Side-by-side HSV-mask coverage is only a visualisation. The health decision still comes from Layer 1, not from which overlay is redder.



c21 09

11. Layer 1 numbers on this sample

These whole-leaf fractions are what the simulation (and the trained health model, in spirit) use. Green dominates; brown is near zero — so the leaf is healthy even when HSV marks some vein pixels.

Signal	Value	How to read it
Green fraction of leaf	92.6%	High → healthy tissue
Brown fraction	8.4%	High → necrotic / brown-spot like
Yellow fraction	2.8%	Streaks / chlorosis
Gray fraction	5.0%	Blast-like gray centres
HSV lesion coverage	28.5%	Teaching overlay only (veins inflate it)
Diagnosis	HEALTHY	production model
Confidence	97.7%	P(healthy) vs P(diseased)
P(healthy)	97.7%	Layer 1 class probability
P(diseased)	2.3%	Layer 1 class probability

12. What this notebook is — and is not

- It **is** a step-by-step documentary of Layer 1 health identification, with the same graphs as the .ipynb.
- It **is** aligned with the Django upload path: preprocess → health → (disease only if diseased).
- It is **not** a replacement for the trained .pkl models in production. The web app still loads cnn_health / rf_health when the environment can.
- HSV red overlays in Step 4 are for teaching. Do not treat high overlay coverage as a disease call on a green leaf.

Re-generate this PDF after changing the notebook:
`python notebooks/build_healthy_leaf_pdf.py`